

Explain to Me

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Intent

Help the learner understand the question and its conceptual purpose, not complete the assignment.

Instructions

When explicitly invoked to explain a homework question:

1. Read the entire question first. Restate its task in plain language and identify what a successful response must address.
2. Explain the important concepts and relationships in the prompt without doing the requested calculations or writing the final answer. Use concrete examples, preferably from social science (especially child development, family studies, or education).
3. Explain why the question is being asked: its underlying learning goal, what the learner should take away, and what kind of reasoning it is practicing.
4. Connect the concepts to the learner's child-development, family-studies, and education interests. Note possible applications to research questions, measurement, evidence, or interpretation.
5. Add other useful conceptual cautions or background the learner needs. Distinguish interpretation from computation, and ask a brief follow-up question if the prompt is ambiguous.

Do not provide the solution, hidden calculations, or a submission-ready response. End with a short checklist of ideas the learner can use to solve the question independently.

Homework 03 Solutions

Question 2: Ridge regression with nearly collinear predictors

R

► Show the solution code

```
correlation    largest_singular_value    smallest_singular_value
      0.9977                14.1415                0.4754
```

► Show the solution code

Empirical mean (standard deviation) across 200 repetitions

lambda	beta1	beta2	sum	difference
0.00	1.496 (1.544)	1.510 (1.549)	3.005 (0.095)	-0.014 (3.091)
0.02	1.487 (0.162)	1.489 (0.166)	2.976 (0.094)	-0.002 (0.314)
0.20	1.368 (0.046)	1.369 (0.047)	2.737 (0.086)	-0.001 (0.034)

► Show the solution code

Average training and test MSE

lambda	training_mse	test_mse
0.00	0.965	1.062
0.02	0.975	1.049
0.20	1.077	1.133

Question 3: What does ridge regression shrink?

Solution

Differentiate the objective with respect to β :

$$\nabla L_{\lambda}(\beta) = -\frac{1}{n}\mathbf{X}^T(\tilde{\mathbf{y}} - \mathbf{X}\beta) + \lambda\beta.$$

Setting the gradient equal to zero gives

$$\frac{1}{n}\mathbf{X}^T\mathbf{X}\hat{\beta}_{\lambda} + \lambda\hat{\beta}_{\lambda} = \frac{1}{n}\mathbf{X}^T\tilde{\mathbf{y}}.$$

Multiplying by n gives the stated normal equation and hence

$$\hat{\beta}_{\lambda} = (\mathbf{X}^T\mathbf{X} + n\lambda\mathbf{I}_p)^{-1}\mathbf{X}^T\tilde{\mathbf{y}}.$$

For every nonzero vector $\mathbf{a} \in \mathbb{R}^p$,

$$\mathbf{a}^T (\mathbf{X}^T \mathbf{X} + n\lambda \mathbf{I}_p) \mathbf{a} = \|\mathbf{X}\mathbf{a}\|_2^2 + n\lambda \|\mathbf{a}\|_2^2 > 0$$

when $\lambda > 0$. The matrix is therefore positive definite and invertible, even when $\mathbf{X}^T \mathbf{X}$ is singular.

Using the singular value decomposition,

$$\hat{\boldsymbol{\beta}}_\lambda = \mathbf{V}_r (\mathbf{D}_r^2 + n\lambda \mathbf{I}_r)^{-1} \mathbf{D}_r \mathbf{U}_r^T \tilde{\mathbf{y}}.$$

If $r < p$, directions orthogonal to the columns of $\mathbf{V}_r$ lie in the null space of $\mathbf{X}$. Ridge sets those coefficient components to zero because they increase the penalty without changing the fitted response.

Multiplying by $\mathbf{X} = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r^T$ gives

$$\mathbf{X} \hat{\boldsymbol{\beta}}_\lambda = \sum_{j=1}^r \frac{d_j^2}{d_j^2 + n\lambda} \mathbf{u}_j \mathbf{u}_j^T \tilde{\mathbf{y}}.$$

Thus, the fitted response component in direction $\mathbf{u}_j$ is multiplied by $\rho_j(\lambda)$. Here $n\lambda = 1$, so

$$\begin{aligned} \rho_1 &= \frac{100}{101} \approx 0.9901, \\ \rho_2 &= \frac{4}{5} = 0.8, \\ \rho_3 &= \frac{0.04}{1.04} \approx 0.0385. \end{aligned}$$

Including the intercept,

$$\text{df}_{\text{eff}}(0.01) = 1 + 0.9901 + 0.8 + 0.0385 \approx 2.8286.$$

The third direction receives the strongest shrinkage. Its small singular value means that the data provide little information in that direction, so an unregularized estimate can have high variance. Shrinkage reduces this sensitivity to response noise. However, if the true mean response has a large component in the third left singular-vector direction, reducing that component to about 3.85% of its least-squares value can introduce substantial prediction bias.

Question 4: Ridge regression and optimization

Solution

The condition number is the ratio of the largest to the smallest eigenvalue. Without ridge,

$$\kappa(\mathbf{A}_0) = \frac{9}{0.01} = 900.$$

Adding $\lambda = 0.09$ changes the two eigenvalues to

$$9 + 0.09 = 9.09 \quad \text{and} \quad 0.01 + 0.09 = 0.10.$$

Therefore,

$$\kappa(\mathbf{A}_\lambda) = \frac{9.09}{0.10} = 90.9.$$

The ridge objective is still more curved in one direction than the other, but the disparity is much smaller. Its contours are less elongated, which improves numerical conditioning.

Let $\widehat{\beta}_\lambda$ denote the ridge minimizer and define

$$\mathbf{e}^{(k)} = \boldsymbol{\beta}^{(k)} - \widehat{\beta}_\lambda.$$

Because the ridge objective is quadratic,

$$\nabla L_\lambda(\boldsymbol{\beta}^{(k)}) = \mathbf{A}_\lambda \mathbf{e}^{(k)}.$$

The error update is consequently

$$\mathbf{e}^{(k+1)} = (\mathbf{I}_p - \eta \mathbf{A}_\lambda) \mathbf{e}^{(k)}.$$

In an eigendirection whose unpenalized eigenvalue is a , $\mathbf{A}_\lambda$ has eigenvalue $a + \lambda$. The corresponding error component is therefore multiplied by $1 - \eta(a + \lambda)$.

Every direction contracts when

$$|1 - \eta(a + \lambda)| < 1$$

for every eigenvalue. It is enough to impose

$$0 < \eta < \frac{2}{M} = \frac{2}{9.09} \approx 0.2200.$$

For $\eta = 1/M$, the contraction factors are

$$1 - \frac{9.09}{9.09} = 0$$

in the strong direction and

$$1 - \frac{0.10}{9.09} \approx 0.9890$$

in the weak direction. The strong-direction error disappears in one step, but the weak-direction error decreases slowly.

Starting with a unit error component in each direction, their magnitudes after one iteration are therefore 0 and approximately 0.9890. After three iterations they are

$$0 \quad \text{and} \quad \left(1 - \frac{0.10}{9.09}\right)^3 \approx 0.9674.$$

For $\eta = 2.1/M$, the strong-direction factor is

$$1 - \frac{2.1(9.09)}{9.09} = -1.1.$$

Its magnitude exceeds one, so a nonzero initial error component in that eigendirection alternates sign and grows. Under the condition stated in the question, the method does not converge even though the weak-direction component may still contract.

The penalty defines the statistical estimator. A larger value can reduce variance and improve conditioning, but it also introduces more shrinkage bias. The step size and numerical algorithm should be chosen to solve the objective accurately. The penalty should be chosen using statistical information about prediction, such as cross-validation or GCV, rather than computational speed.

Question 5: Selecting the ridge penalty for real-estate prediction

Solution

The ridge fitting function below performs three statistical operations. It estimates the predictor means and scales from the fitting observations, solves the ridge normal equation on the standardized scale, and transforms the coefficients back to the original predictor scale for prediction. It is used for the final refits. In the cross-validation loop, each fold's means, scales, and response center are computed once and reused across every penalty. In the GCV loop, the full training means, scales, and response center are likewise computed once and reused across the full penalty grid.

Write $\bar{x}_j$ and s_j for the fitting-sample mean and scale of raw predictor j , and $\bar{y}$ for the fitting-sample response mean. The code uses `x_bar`, `s`, and `y_bar` for these quantities, `X` for standardized predictors, and `beta_hat` for their fitted slopes. Conversion to raw-scale coefficients is

$$\widehat{\beta}_{\text{raw},j} = \frac{\widehat{\beta}_j}{s_j}, \quad \widehat{\beta}_{0,\text{raw}} = \bar{y} - \sum_{j=1}^p \bar{x}_j \widehat{\beta}_{\text{raw},j}.$$

The code calls these `beta_hat_raw` and `raw_intercept`. Both representations give the same predictions.

For GCV, it is unnecessary to construct the full $n \times n$ smoother matrix. The residual vector gives $(\mathbf{I}_n - \mathbf{H}_\lambda)\mathbf{y}$, and the singular values give

$$\text{tr}(\mathbf{H}_\lambda) = 1 + \sum_{j=1}^p \frac{d_j^2}{d_j^2 + n\lambda}.$$

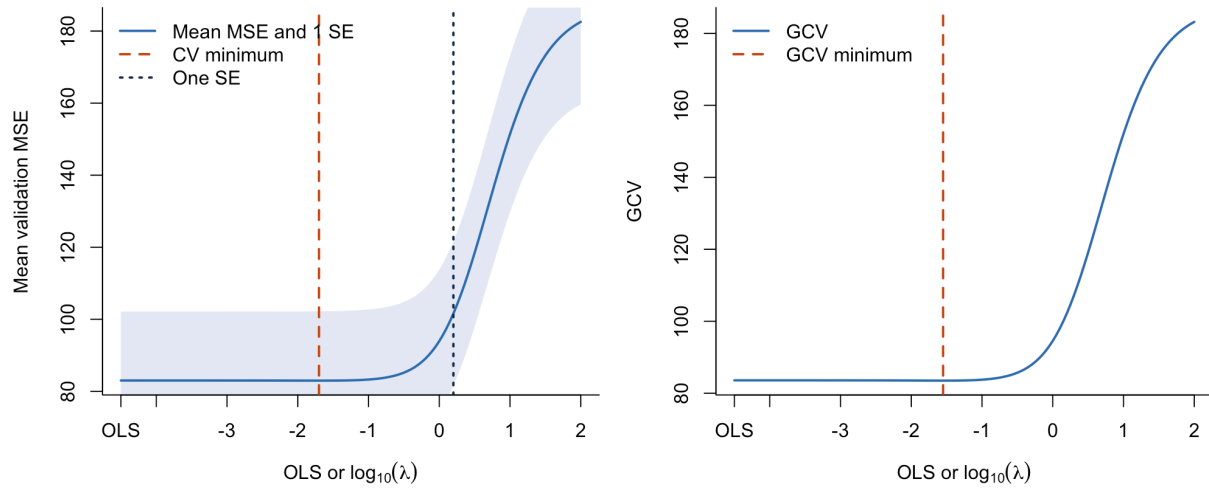
This is algebraically equivalent and avoids storing a large matrix.

The quantity called `cv_se` below is the conventional one-standard-error summary across folds. Because the ten fitted training sets overlap, it is a useful model-selection heuristic, not an independent-sample standard error.

R

► Show the solution code

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Training-only penalty selection by ten-fold cross-validation and GCV.

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```
lambda_min    lambda_1se    lambda_gcv    one_se_limit
0.0199526     1.5848932     0.0281838    102.2075631
```

► Show the solution code

Final training and test comparison

rule	lambda	df_eff	train_mse	test_mse
OLS	0.00000	7.000	80.094	66.322
CV minimum	0.01995	6.773	80.135	66.000
One SE	1.58489	3.033	100.374	81.200
GCV	0.02818	6.689	80.171	65.903